Industrial Marketing Management 63 (2017) 116–128

Contents lists available at ScienceDirect

Industrial Marketing Management

A diagnostic model of private control and collective control in buyer-supplier relationships

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ARTICLE INFO

Article history:
Received 10 September 2015
Received in revised form 12 October 2016
Accepted 28 November 2016
Available online 4 December 2016

Keywords:
Buyer-supplier relationships
Control
Governance
Interorganizational exchange

ABSTRACT

This study examines the control-based governance in buyer-supplier relationships. Building on boundary spanning theory and governance literature, we propose an integrated model that consists of exchange parties' private control (aimed at individual gains) and collective control (aimed at joint gains) in boundary spanning activities, along with their structural antecedents and relationship consequences in interorganizational governance. Using data collected from manufacturer-distributor dyads, we demonstrate that a buyer-supplier relationship characterized by a high degree of distributive justice and low degrees of goal difference and power asymmetry promotes exchange parties' collective control while inhibiting private control in boundary spanning conduct. The impact of private and collective controls on dyadic relationship performance is further mediated through governance costs and returns. Specifically, private control results in conflict and transaction costs that undermine dyadic relationship performance, whereas collective control leads to solidarity and reciprocity that sustain dyadic relationship performance. Recognizing and distinguishing between private control and collective control is essential to managing boundary-spanning behavior in buyer-supplier relationships and to solidifying relationship performance in supply chain and channel management.

1. Introduction

The abundant research on interorganizational relationships in the past few decades has clearly documented the role of collaboration in value creation as well as the importance of control in interorganizational governance (Bensaou & Venkatraman, 1995; Jap & Ganesan, 2000; Majchrzak, Jarvenpaa, & Bagherzadeh, 2015). Control, a mode of structuring and regulating the conduct of parties, is essential to achieving firm performance and managing interorganizational relationships (Das, 1989; Das & Teng, 1998; Luo, 2007; Williamson, 1979), because proper control can curtail the risk and uncertainty that are inherent in exchange (Claycomb & Frankwick, 2010; Larson, 1992). Thus, choosing proper control practices is key to interorganizational relationships management (Hoetker & Mellewigt, 2009) because "coordination and other desirable outcomes do not occur unless members recognize and choose behaviors most efficient and effective for the conduct" (Gilliland, Bello, & Gundlach, 2010, p. 441). Accordingly, much research has been devoted to investigating control in interorganizational governance and results have been prolific (e.g., Das & Teng, 1998; Gilliland et al., 2010; Heide, Wathne, & Rokkan, 2007; Jap & Ganesan, 2000; Varoutsa & Scapens, 2015).

To date, control[1] has been studied in various ways, including the degree of importance and effectiveness (primary & secondary controls; see Jaworski, 1988), point of intervention (process & output controls; see Jaworski, 1988), ownership structure (equity & non-equity controls; see Yan & Gray, 1994), control dimension (incenting, monitoring, & enforcing, see Gilliland et al., 2010), governance mechanism (transactional/formal & relational/social, see Popp & Zenger, 2002), and parties involved (private & collective controls, see Luo, 2007). Despite their fruition of providing multiple facets of the control practices, many of prior control typologies focus on control exerted by a single party. An

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[1] We distinguish *control* from *governance*. *Governance* is regarded as "a higher-level concept describing an organizational construction or, in broader term, institutional framework" whereas *control* is "the underlying and concrete operative practice" that is brought to achieve the desirable or predetermined goal (Hoetker & Mellewigt, 2009, p. 1027). Different control activities (e.g., process and output controls) may be adopted by two parties to support interorganizational governance (e.g., a contractual arrangement).

http://dx.doi.org/10.1016/j.indmarman.2016.11.003
0019-8501/Published by Elsevier Inc.

exception is the private-collective control scheme introduced for parent firms in the context of joint ventures and strategic alliances (Luo, 2007; Luo, Shenkar, & Gurnani, 2008), where private control is used by one party to maximize individual gain at the expense of the partner and collective control is employed by both parties to pursue joint governance for the benefit of both. This control scheme explicitly separates private control exerted by individual parties and collective control by both parties so that both types of controls can be examined in horizontal interorganizational relationships.

In the context of buyer-supplier relationships (i.e., vertical interorganizational relationships), although the governance literature has evolved to identify contractual governance (based on formal, unilateral, hierarchical authority) and relational governance (based on informal, bilateral, social interactions) as major governance mechanisms[2] (Heide, 1994; Poppo & Zenger, 2002), prior studies fail to recognize that unilateral and collective controls may simultaneously present in a buyer-supplier relationship (BSR) under both types of governance. For example, under contractual governance, the buyer can choose to hide the market intelligence from the supplier (i.e., private control) when processing information or work together with the supplier to determine the market demand (i.e., collective control). Because a BSR is a loosely coupled system in which two parties need to collaborate in carrying out supply chain/channel functions while having separate goals and interests performing boundary spanning roles (Aldrich & Herker, 1977; Liu, Huang, Luo, & Zhao, 2012), an understanding of private and collective controls in interparty boundary spanning interfaces and their impacts on BSRs is important to effective supply chain/channel management. As such, we are interested in diagnosing private and collective controls that are inherent in BSRs.

Our study contributes to the control/governance literature by extending and examining private and collective controls under interorganizational governance in BSRs. This alternative control typology adds to other control practices and provides a useful angle to examining boundary spanning conduct between the buyer and the supplier. Further, we depict a more complete picture of the control model by identifying the structural antecedents to and the performance consequences of private and collective controls in BSRs. With the integrated model including the effects of both private control and collective control on dyadic relationship performance (compared to single party performance in many previous studies), we seek to provide meaningful guidance on effective supply chain/channel management because (1) many managerial decisions (e.g., partner selection decisions) are partly driven by the underlying characteristics of the dyadic links in the exchange relationship (Majchrzak et al., 2015), and (2) the longevity of interorganizational collaborations in supply chains and distribution channels is contingent on the fact that both parties need to benefit from the partnership (Dyer, 1997).

In summary, the intent of this study is three-fold: (1) to examine private and collective controls in interpary boundary spanning conduct; (2) to explore structural conditions of a BSR that may stimulate private and collective controls; (3) to elucidate the various consequences of private and collective controls in a BSR. By examining the two distinct types of controls and establishing paths through which how private and collective controls are determined by relationship structural conditions, generate governance consequences, and impact dyadic relationship performance, we hope to add insights to a more nuanced understanding of controls in interorganizational governance.

[2] Heide (1994) proposes a typology of governance: market, unilateral/hierarchical (e.g., franchising contracts), and bilateral governances. Authority-based, unilateral governance and relationship-based, bilateral governance have evolved to contractual governance and relational governance respectively. Collective control should not be taken as equivalent to relational governance because one party may unilaterally seek for developing social ties while the other party may refrain from such connections (see Huang, Sternquist, Zhang, & Calantone, 2011). Private control and collective control may coexist under either contractual governance or relational governance.

2. Theoretical framework and hypotheses development

2.1. A private-collective control scheme

Control refers to modes, mechanisms, or processes by which exchange parties structure and regulate their activities and behaviors (Das & Teng, 1998; Luo, 2007; Williamson, 1979). As illustrated by Das and Teng (1998, p. 493), "the purpose of control is to fashion activities in accordance with expectations so that the ultimate goals of the organization can be attained". Specifically, control is recognized as the underlying operative practices which are brought to bear between the two parties or actual activities exchange parties can deploy in order to achieve desired outcomes (Hoetker & Mellewigt, 2009; Jaworski, 1988). Relationship control, risk minimization, and performance improvement are major concerns in various types of interorganizational collaboration (Gan, Sethi, & Yan, 2005; Langfield-Smith & Greenwood, 1998).

In the context of joint ventures (JV) and strategic alliances (SA), Luo (2007) and Luo et al. (2008) propose the classification of parent control of the JV or the SA into private control and collective control. According to Luo (2007), private control refers to unilateral actions taken by one party to ensure that the JV or the SA is managed to maximize its own benefit, often taking advantage of the shared resources, whereas collective control refers to bilateral actions jointly designed and taken by both parties to guide, monitor, and oversee the JV or the SA in order to ensure mutual gain for both parties from collaboration. Thus, private control and collective control differ in nature (unilateral vs. bilateral), underlying intent (self-interest vs. joint interests), and visibility (mostly covert vs. always overt) (Luo, 2007). Because private control implies self-interest at the expense of the other party, whereas collective control signifies joint actions for mutual gain, the two types of control seem to have different effects on the partnership (Luo et al., 2008). Applying the private-collective control scheme to the context of BSRs, a major vertical form of interorganizational relationships in supply chains and distribution channels, we propose that private control and collective control are highly relevant and manifested in boundary spanning activities between the buyer and the supplier. See Fig. 1 for the conceptual model.

2.2. Private and collective control in buyer-supplier relationships

According to boundary spanning theory (Aldrich & Herker, 1977), organizations depend on collaborations with others for critical resources in order to survive and grow, and boundary spanners between organizations facilitate the roles of information processing and external representation in interorganizational collaboration. As a typical interorganizational collaboration in supply chains and distribution channels, a BSR can be viewed as a loosely coupled system in which the buyer and the supplier are separate and independent yet connected and interdependent (Majchrzak et al., 2015; Orton & Weick, 1990). The buyer and the supplier independently and interdependently perform their boundary spanning roles of information processing (e.g., exchanging sales forecasts and inventory data, sharing proprietary product designs) and external representation (e.g., negotiating contracts, resolving conflicts) in a BSR. Such a loosely coupled system allows the buyer and the supplier to maintain and serve their individual needs (loose) while collaborating with each other to achieve joint goals (coupled) through boundary spanning activities (Aldrich & Herker, 1977; Stock, 2006). Meanwhile, the coexistence of looseness and coupling provides impetus for the buyer and the supplier to engage in private and collective controls, with private control aiming at individual gain at the expense of the other party (loose) and collective control striving for joint interest (coupled) (Luo, 2007). Therefore, a BSR is both loose and coupled, yielding simultaneously private and collective controls.

As an illustration of private control and collective control, we use an original equipment manufacturer (the buyer) and a contract manufacturer (the supplier) to showcase private and collective controls of demand forecast coordination in interparty boundary spanning conduct

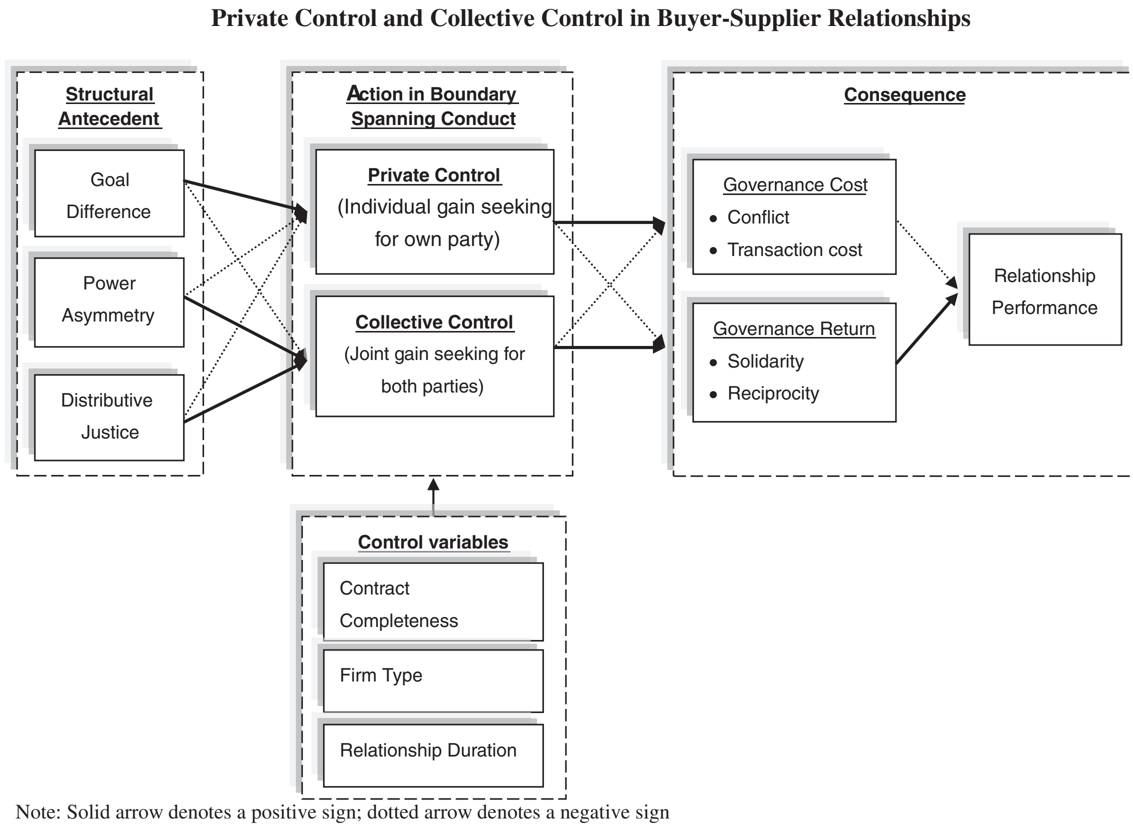

Fig. 1. Private control and collective control in buyer-supplier relationships. Note: Solid arrow denotes a positive sign; dotted arrow denotes a negative sign.

(i.e., information processing), as noted in Thomas, Warsing, and Zhang (2009). The buyer purchases subassemblies provided by the supplier that are made from two complementary sets of components—one needs a long lead-time and the other needs a short lead-time. At the supplier's request, the buyer shares a demand forecast with the supplier when the supplier needs to order the long lead-time set of components, and also shares forecast updates afterward. This practice demonstrates collective control between the buyer and the supplier. Alternatively, the buyer may unilaterally decide not to provide forecast updates, which affects the supplier's ordering of the short lead-time set of components, so that the supplier has to bear the full overage costs. This practice signifies the buyer's private control which is imposed to purposefully avoid sharing the overage costs with the supplier (Thomas et al., 2009). From this illustration, it is clear that, in a BSR, one party can choose to wield private control or to engage in collective control when performing boundary spanning functions, depending on the needs whether to maximize its own gain (at the expense of the other party, as revealed in Thomas et al., 2009) or to realize shared goals for mutual benefit. Thus, in the context of BSRs, we define private control as *unilateral actions taken by one party who seeks to maximize individual gain at the expense of the other party* and collective control as *joint actions exercised by both parties for mutual benefit* in conducting boundary spanning activities.

Further, we submit that private and collective controls may be *simultaneously* present in a BSR because the two parties need to cooperate in carrying out boundary spanning activities (Marrone, Tesluk, & Carson, 2007), yet they may have separate identities and individual goals when performing their boundary spanning roles. In any BSR, there will always be some degree of both private and collective control of boundary spanning behavior. For example, in order to reduce the bullwhip effect and minimize inventory costs, supply chain partners openly share accurate inventory information with each other—a practice of collective control on information processing. Other collective control between partners may include such external representation behaviors as resource sharing, contract stipulation, operation process coordination, new product co-development, joint problem solving, etc. (Majchrzak et al., 2015; Stock, 2006). Meanwhile, because both the buyer and the supplier are independent entities with their own organizational cultures, strategic goals, and operational processes, they may engage in unilateral tactics in pursuit of self-interest when performing information processing (e.g., holding back key information; hiding core technologies) and external representation (e.g., changing prices; manipulating contracts; terminating relationships) functions (Georgiades, 2008; Jap & Anderson, 2003). For instance, for many years the Carrefour Group (a France-based multinational retailer and the second largest retailer in the world) has been proud of its supply chain in China. Collaborating with its suppliers, Carrefour has successfully managed an efficient supply chain for its >230 stores in 70 Chinese cities as of 2015. However, its suppliers have recently accused Carrefour of unilaterally adding new surcharges such as store grand-opening fees, anniversary fees, and holiday celebration fees (i.e., private control), in addition to the typical slotting fees agreed upon by both parties (i.e., collective control). Carrefour's unilateral actions have imposed a significant burden on its suppliers, and have caused many suppliers to end contracts and to terminate their relationships with Carrefour. For example, Master Kang, a top instant noodle manufacturer in China, has recently broken off with Carrefour (Xinhua News, 2013). The Carrefour case demonstrates that both private control and collective control may co-exist in a BSR, and that they may have different effects on the exchange relationship. In the next sections, we further identify the structural conditions that breed private and collective controls, as well as the varying consequences of these controls in the dyadic links of BSRs.

2.2.1. Structural conditions of private and collective control in BSRs

As aforementioned, boundary spanning theory predicts how organizations and their boundary spanning representatives, with resource scarcity and increasingly close linkages, coordinate to achieve prearranged goals while safeguarding each party's own interests (Aldrich & Herker, 1977; Jap & Ganesan, 2000) or protecting itself from disruptive forces (Hibbard, Kumar, & Stern, 2001). As such, parties may choose to either jointly facilitate boundary spanning activities (this involves collective control) or engage in competitive behavior (this involves private control) to maximize own interests (Luo et al., 2008; Marrone et al., 2007). According to boundary spanning theory, an organization's actions in boundary spanning roles are determined, in part, by the nature of its interface with the other organization (Aldrich & Herker, 1977; Luke, Begun, & Pointer, 1989), and the nature of the boundary spanning interfaces refers to the positions and dynamics between organizations (Luo, 2007; Perry & Angle, 1979). Following this logic, we submit that the control practices of the buyer and the supplier over their boundary spanning behaviors are partially determined by the interface between the two parties. Drawing from previous BSRs studies in supply chain/channel literature (e.g., Gaski, 1984; Gilliland et al., 2010; Griffith, Harvey, & Lusch, 2006; Jap, 2001; Kumar, Scheer, & Steenkamp, 1995a, 1995b; Liu et al., 2012; Yan & Gray, 1994), we identify three structural conditions that characterize a BSR—*goal incongruence* (representing interparty value position), *power asymmetry* (representing interparty resources position), and *distributive justice*[3] (representing interparty equity position) to represent the boundary spanning interfaces in a BSR. These three structural conditions are also consistent with the dynamics identified in interorganizational collaboration (Hingley, 2005; Majchrzak et al., 2015).

2.2.1.1. Goal incongruence. A goal is the anticipated objective and the direction toward which an organization's endeavor is guided (Rossetti & Choi, 2008). In a BSR, when the buyer and the supplier have similar perceived values and expectations of the exchange relationship, they have goal congruence (Rossetti & Choi, 2008). In such an instance, two parties have closely aligned expectations and their objectives are similarly intertwined; consequently, they believe that their interests can be best realized through collective decisions and actions in boundary spanning activities (Dyer, 1997; Marrone et al., 2007). For instance, the CPFR practice (Collaborative Planning, Forecasting, and Replenishment) in the retail industry was initiated together by Procter & Gamble and Walmart in the 1990s. The collective control of inventory management was a direct result after Procter & Gamble and Walmart decided and committed to forgo their unilateral plots, to trust each other in boundary spanning conduct, and to align their goals together for developing a collaborative relationship (see Kumar, 1996).

On the other hand, when the goals of two parties are mutually exclusive, or are incompatible with or greatly different from each other, a situation of goal incongruence occurs. In this situation, the buyer and the supplier often cannot achieve their respective goals concurrently because of a lack of alignment in value perceptions between them (Jap, 2001). For example, while the buyer desires the lowest purchasing cost, the supplier strives to achieve the highest profit margins or revenues. Goal incongruence between two parties is often attributable to a lack of common interest, leading to unilateral actions driven by self-interest associated with opportunism (Jap, 2001; Williamson, 1979). When parties care only about their own interests, their behaviors in conducting boundary spanning activities that are related to information processing (e.g., not sharing updated forecasts) and external representation (e.g., unilateral actions that defect from contract terms) become difficult to predict. With doubt and suspicion gradually permeating the relationship, parties tend to engage in covert tactics for self-gain and are, therefore, less likely to work together in boundary spanning activities, as in the case of the bitter relationship between Procter & Gamble and Walmart prior to 1980s (see Kumar, 1996). Thus, we propose that goal incongruence between a buyer and a supplier breeds private control and limits collective control in BSRs.

H1. In a buyer-supplier relationship, goal incongruence is positively associated with private control but inversely associated with collective control, ceteris paribus.

2.2.1.2. Power asymmetry. Power asymmetry between two parties refers to an imbalance in resource possession, strategic position, bargaining ability, and perceived importance to the relationship (Cuevas, Julkunen, & Gabrielsson, 2015; Stock, 2006; Yan & Gray, 1994). Power, is the ability of one party to influence the decisions of the other party (Brown, Lusch, & Nicholson, 1995). One organization usually achieves power over another through possession of scarce resources, thus making the other organization dependent (Emerson, 1962). In a BSR, the power of the buyer over the supplier is equal to the dependence of the supplier on the buyer (Emerson, 1962). When the dependence between two parties is unbalanced, power asymmetry occurs (Brown et al., 1995). The relatively more powerful party possesses more resource advantage and has superior authority in the relationship, therefore it is less likely to compromise (Gaski, 1984; Kumar et al., 1995a, 1995b) and more likely to engage in unilateral actions such as ignoring the other party or dealing with the other party using coercion (Kim, 2000). For instance, the engineering construction industry in the UK in the 1980s and 1990s was dominated by large management contractors, and these large organizations often use their dominance to pass costs on to small individual subcontractors (Korczynski, 1996). This so-called "top down" control (Korczynski, 1996, p. 787) represents private control exerted by the dominant party in the BSR with power asymmetry. Similarly, General Motors was reported by its small suppliers for wielding private control in cases of patent infringement, contract breach, and price-cutting pressures over time (Dyer, 1997).

Alternatively, the less powerful party has disadvantages in the BSR and bears more relational risks. To avoid being exploited by the more powerful party, the less powerful one may also take actions through overt or hidden tactics to protect its own interests, rather than seeking joint investment and decision making with the more powerful one, because the less powerful party has a lower level of trust in, and satisfaction with, the other party (Brown et al., 1995). For example, in the same cases elucidated in Korczynski (1996, p. 794), small individual subcontractors were also reported for unilateral actions involving deceitful behaviors such as "sitting on the job" and "withholding information". Similarly, in the case of Carrefour and its suppliers in China, many suppliers chose to end the relationship with Carrefour due to that company's unilateral decision to charge additional promotion fees. In a BSR with high power asymmetry, both parties may be less inclined to adopt collective control, and more inclined to employ private control, in order to pursue (the more powerful party) or protect (the less powerful party) their own interests through information processing (e.g., withholding key information) and external representation (e.g., forcing unreasonable price cutting). Thus, we propose:

H2. In a buyer-supplier relationship, power asymmetry is positively associated with private control, but inversely associated with collective control, ceteris paribus.

2.2.1.3. Distributive justice. Distributive justice refers to the ways in which resources and outcomes are distributed between the two parties, and parties' perceptions of fairness of distribution (Kumar et al., 1995a, 1995b). If the output and reward that the buyer and the supplier obtain

[3] Distributive fairness and procedural fairness are both considered as the structural dimensions of perceived fairness due to their concerns for equity distribution and formal procedures respectively between two parties (Tyler & Bies, 1990). We chose distributive fairness rather than procedural fairness as a structural condition in a BSR because distribution fairness indicates two parties' stances in equity distribution whereas procedural fairness involves two parties' perceptions of formal procedures and processes (Kumar, Scheer & Steenkamp, 1995a, b), which represent the perceived resource distribution in a BSR.

from a BSR properly reflects their respective input and investment in the relationship, they will form perceptions of the relationship being fair and worth maintaining (Kumar et al., 1995a, 1995b), and such perceptions are conveyed to their behaviors and decisions in boundary spanning conduct (Friedman & Podolny, 1992). Or else, if one party perceives a low level of distribution justice, the BSR is likely to dissolve. For instance, as reported in the *Wall Street Journal* (Georgiades, 2008), in 2007 the Walmart Canada Corp. (a unit of retailing giant Wal-Mart Stores, Inc.) requested that the Lego Group lower its prices—higher in Walmart's Canadian stores due to the appreciation of the Canadian dollar—so that the Canadian pricing was aligned with U.S. pricing and the savings could be passed on to consumers, but Lego was reluctant to do so. Having failed to obtain a satisfactory response from Lego Group, Walmart Canada decided to discontinue its business with the Lego Group in 2008. In this case, Walmart Canada viewed Lego's rejection of lowering prices as unfair (i.e., a distributive justice issue) and decided to take unilateral action in performing external representation (i.e., private control) to terminate the relationship.

In contrast, in a BSR with a high level of distributive justice, both parties perceive that their investments are adequately recognized and will be rewarded proportionally; thus, they worry less about the other party's exploitation and opportunism in the relationship (Griffith et al., 2006). Consequently, both buyer and supplier are more willing to maintain the relationship (Kumar et al., 1995a, 1995b), to work with each other in performing boundary spanning roles in the operations process, and to guide the development of the long-term relationship (Griffith et al., 2006). Because of the confidence about their expected gain from the relationship and a belief that the relationship would not be exploited by the other party (Larson, 1992), they are also more willing to work together to maximize their joint gain, recognizing that the maximization of joint gain means the maximization of individual gains (Bensaou & Venkatraman, 1995). As such, the buyer and the supplier in a BSR with a high level of distributive justice are more likely to share control with each other for the sake of mutual gain, and to jointly process boundary spanning activities related to information processing (e.g., information sharing) and external representation (e.g., conflict resolution) in the relationship (Bensaou & Venkatraman, 1995). They are, as well, more likely to refrain from engaging in unilateral self-interest-seeking behaviors in boundary spanning activities that would harm the relationship, as reflected in a collaborative relationship between Toyota Australia and its major carpet supplier Trim, reported in Langfield-Smith and Greenwood (1998). Thus, we propose:

H3. In a buyer-supplier relationship, distributive justice is negatively associated with private control, but positively associated with collective control, ceteris paribus.

2.2.2. Consequences of private and collective control

The ultimate purpose of control in an exchange relationship is to create value and improve performance through relationship management (Jap & Ganesan, 2000). Luo et al. (2008) hinted that private and collective controls may drive multiple consequences. We follow this thought and expect that various consequences emerge sequentially. Drawing from two dominant streams in governance research and based on different natures of private control and collective control, we propose governance cost (based on transaction cost economics) and governance return (based on the relational view) as the immediate governance consequences, and relationship performance as the ultimate outcome of a BSR.

In the studies of transaction cost economics, transaction cost is viewed as a major governance cost[4] (Williamson, 1979), particularly in emerging economies where exchange parties face resource scarcities, market uncertainty, and regulatory variability (Wright, Filatotchev, Hoskisson, & Peng, 2005). Transaction cost in an exchange relationship includes bargaining costs associated with negotiations, monitoring costs derived from executions, and mal-adaptation costs embodied in unforeseen uncertainty (Dyer, 1997; Williamson, 1979). Conflict, as one of the most potentially destructive outcomes in BSRs (Palmatier, Dant, & Grewal, 2007), is often documented in boundary spanning conduct between the buyer and the supplier (Gaski, 1984; Jap & Ganesan, 2000). Due to failures in communication (i.e., information processing) and coordination (i.e., external representation) (Aldrich & Herker, 1977) between boundary spanners in carrying out supply chain/channel activities, conflict may rise (Jap & Anderson, 2003).

In contrast to conflict and transaction costs that are typically treated as governance cost (Heide et al., 2007), solidarity and reciprocity[5] are regarded as governance return from a collaborative relationship from a relational governance perspective (Kim, 2000). Solidarity refers to the behavioral orientation of parties (Kim, 2000), and reciprocity is a pattern of responsive behaviors between two parties (i.e., A's positive behavior toward B is driven by or is in response to B's positive behavior toward A) (Johnson & Sohi, 2001). Both solidarity and reciprocity imply dyadic relationship status resulting from a collaborative relationship, which nurtures two-way communication (i.e., information processing) and coordination (i.e., external representation) in boundary spanning activities (Anderson & Narus, 1990).

We use relationship performance in referring relationship performance in a BSR and define the term as the relationship-specific performance jointed achieved by the buyer and the supplier (Palmatier et al., 2007). The ultimate goal of a successful relationship is to achieve relationship performance because a performing partnership means that both parties achieve benefits that include financial, operational, and overall effectiveness performance (Geyskens & Steenkamp, 2000); otherwise, the partnership is not likely to be stable and sustainable. Relationship performance of an exchange relationship, however, is affected by governance cost and return incurred in the relationship. Next we describe how interparty private and collective controls influence relationship performance through governance cost and return.

The intent of private control is to maximize individual gain at the expense of the other party, or even at the expense of possible joint gain (Luo et al., 2008). This tactic makes it difficult for the buyer and the supplier to perform proper boundary spanning functions such as information sharing and processing (Aldrich & Herker, 1977), because sharing key information such as strategic pricing, advance order information, and updated demand forecasts may make one party vulnerable to the other party's opportunism (Majchrzak et al., 2015). Thus, private control induces hostility and causes conflict in the relationship (Luo et al., 2008). The intensified degree of conflict further reduces motivation of parties when they each perform their external representation role in boundary spanning activities (Anderson & Narus, 1990). As a result, the exchange relationship will neither function nor meet desired goals, and the overall relationship performance may suffer (Griffith et al., 2006). As well, private control in boundary spanning conduct implies higher transaction costs in the external representation function (Aldrich & Herker, 1977). Because most private controls are covert tactics, parties have to constantly monitor each other in order to detect hidden opportunistic behaviors (Heide et al., 2007). When opportunism does occur, parties must take actions such as investigations, negotiations, conflict resolution, and further decision making about the relationship, thus generating additional adaptation costs. As an example, a buyer's decision not to share updated forecasts will affect the supplier's short lead-time purchase decisions, and the time the supplier waits for updates affects its long lead-time purchase decisions (Thomas et al., 2009), generating

[4] Opportunism is also a central concept in transaction cost economics. Opportunism, however, as implied in Williamson (1979), is often seen as a cause for the need to control and monitor. In Palmatier et al. (2007), opportunism is theorized as an antecedent in inter-organizational relationships.

[5] Trust and commitment are two other key concepts in relational governance studies. However, these two are often seen as preconditions for collaboration to happen, as illustrated in the case of Walmart and P&G before.

additional transaction costs for the supplier. Higher transaction cost, in turn, increases expenditures and reduces operational efficiency, thus diminishing the overall relationship performance (Williamson, 1979). Because parties practicing private control are concerned solely about their own gain, they are unlikely to support or provide help for each other, thus, the level of solidarity and reciprocity in the relationship is low. Lack of solidarity and reciprocity may further hinder knowledge transfer and resource sharing (Heide et al., 2007), thereby reducing operational efficiency and market competitiveness of the distribution channel or supply chain. Thus:

H4. In a buyer-supplier relationship, private control is negatively associated with relationship performance through increased governance cost (including conflict and transaction cost) and reduced governance return (including solidarity and reciprocity).

In contrast, interparty collective control pursues the maximization of joint gain. Thus, both parties are more likely to coordinate in boundary spanning activities and to be accommodating when unexpected events occur (Gan et al., 2005). The cooperation between the buyer and the supplier such as information sharing help reduce potential conflict (Griffith et al., 2006). Because collective control is jointly designed by both parties via formalized policies, rules, and norms, and is transparently enforced at all hierarchical levels, the responsibilities and obligations for each side are clearly defined and easily monitored under the guiding principle of the maximization of joint gain. Accordingly, negotiation, monitoring, and mal-adaptation costs in the relationship are low; and low transaction costs contribute to partner profit and operational efficiency, thus improving relationship performance. Dyer (1997) presents an in-depth case illustrating how collective actions taken by Japanese automakers (Toyota and Nissan) and their suppliers help reduce transaction costs and increase transaction value in the BSRs.

Further, parties practicing collective control seek to align their role positions and behaviors, holding a common belief that they should support each other (Gan et al., 2005). A relationship characterized by bilateral convergence is likely to nurture norms of solidarity between two parties (Kumar et al., 1995a, 1995b). Thus, parties involved in collective control build a sense of solidarity; when unforeseen circumstances occur, solidarity will enable two parties to support each other and resolve the emerging issues with operational efficiency (Thomas et al., 2009). Under collective control, both parties are confident and are willing to share key information and resources, to integrate operational processes, and to commit to the relationship because they have faith that their good will and relationship investment will be reciprocated (Dyer, 1997). A high level of reciprocity in the relationship leads to interdependence (Heide et al., 2007). Research in supply chain/channel management suggests that social norms such as solidarity, mutuality, and reciprocity reflect the shared expectations and beliefs within dyads, and influence supply chain performance and channel outcomes (Heide et al., 2007; Jap & Ganesan, 2000). Hence:

H5. In a buyer-supplier relationship, collective control is positively associated with relationship performance through reducing governance cost (including conflict and transaction cost) and increasing governance return (including solidarity and reciprocity).

3. Research methods

This study examines private and collective controls in as well as their structural conditions and relationship outcomes in the dyadic links of BSRs. Since the unit of analysis is the buyer-supplier dyad, we collected dyadic data from manufacturer-distributor dyads in the household appliances industry in China to fill the lacuna of the inadequate existing management research on Asia. We selected China as the study venue also because of (1) the vital role played by China in global supply chains, (2) the rapid economic growth of China as the largest emerging market, (3) the sheer size of the Chinese consumer market, which has rendered China as one of the most important powerhouses in the world economy. We chose the household appliances industry because it is one of the most developed and established industries in China, including top brand manufacturers and major distributors featuring long-term collaboration yet that conflicts between the buyers and the suppliers in this industry have been frequently reported by Chinese media, which enable us to study both private and collective controls in the same context. In addition, using data from a single industry helps eliminate industry-level differences in dyadic BSR conditions and business practices.

3.1. Data collection

Significant efforts were directed to collecting dyadic data through the following steps. First, we developed paired questionnaires in English for both manufacturers (suppliers) and distributors (buyers). Back-translation was applied to complete the Chinese version of the questionnaires, and we then used interviews as a way to check the questionnaire with 16 manufacturer-distributor dyads. This pilot test revealed the need for some wording modifications and refining, resulting in the final questionnaires. Second, we obtained a list of 900 nationwide suppliers from the Haier Group, the third largest household appliance manufacturer in the word (the largest in China), and then called these supplier companies, asking whether the marketing manager or director would participate in our survey. We then mailed the coded questionnaire to each of each of these respondents, requesting that he or she select a major distributor of his or her company and complete the survey with regard to the specific relationship with that distributor (i.e. the buyer). A major distributor was defined in the questionnaire as the one to which the supplier sold the largest volume during the last three months. With three rounds of reminders (phone calls and emails), we received 251 completed questionnaires from the suppliers. Finally, we sent the paired questionnaires to the managers of the designated distributors selected by the suppliers. After the same rounds of reminders and eventually sending out doctoral students to collect questionnaires on site, we received 225 completed questionnaires from the distributors (i.e., the buyers).

In addition, to warrant proper respondents, we specifically requested for both surveys that the questionnaires be answered by key informants who are familiar with the focal relationship. As a result, our respondents were senior managers or directors who were responsible for the focal relationship, and the held positions such as vice president of sales, sales manager, merchandise director, and purchasing manager. As an added verification, we included two items in the questionnaires to measure the respondents' qualifications: (1) "How much do you know about your company's dealing with this buyer/supplier?" and (2) "How much have you been involved in your company's dealing with this buyer/supplier?" Both items were based on a five-point Likert scale (1 = "very low" and 5 = "very high"). The mean knowledge scores were 4.23 (SD = 0.56) in the supplier sample and 4.25 (SD = 0.62) in the buyer sample. On average, the buyer-side informants had worked in their current positions for 4.7 years, while the supplier side indicated 3.5 years. Further, to reduce response bias in self-reporting, we rearranged the order of the sections to create different versions of the questionnaire (Podsakoff, Mackenzie, Podsakoff, & Lee, 2003). Finally, we checked non-response bias by randomly choosing 50 firms that returned their questionnaires, and another 50 firms that neither completed nor returned their questionnaires and collected firm information such as firm size, sales volume, location, ownership, and relationship duration for a comparison *t*-test. All t-statistics were non-significant. Thus, non-response bias is not a threat in our data.

3.2. Measurement

Table 1 and Table 2 show the constructs and descriptive statistics. All key constructs (see Table 1) were measured by multiple items on 7-

Table 1
Reliability and Validity.

	Cronbach's α			Factor loading			CR			AVE			Rwg
	B	S	BS	B	S	BS	B	S	BS	B	S	BS	
Goal difference (adapted from Luo, 2002)	0.96	0.96	0.96				0.97	0.97	0.97	0.89	0.89	0.90	0.896
Both sides have different opinions on risk minimization.				0.92	0.92	0.93							
Both sides have different opinions on maintaining relationship closeness.				0.96	0.96	0.97							
Both sides have different opinions on maintaining relationship longevity.				0.96	0.95	0.96							
Both sides have different opinions on minimizing sales costs.				0.93	0.93	0.94							
Power (adapted from Brown et al., 1995)	0.76	0.69	n/a				0.86	0.83	n/a	0.68	0.62	n/a	n/a
We may cancel or refuse to renew the contract if the supplier (buyer) does not do what we ask.				0.86	0.85	0.81							
The supplier (buyer) could get some needed help from us by agreeing to our requests.				0.87	0.77	0.78							
We often hint that we would take actions that would reduce the supplier's (buyer's) profits if they do not go along with our requests.				0.74	0.74	0.62							
Distributive Justice (adapted from Kumar, Scheer, & Steenkamp 1995)	0.91	0.91	0.92				0.94	0.94	0.95	0.85	0.85	0.86	0.918
Our rewards or returns are fair in view of our resource contribution to the relationship				0.92	0.93	0.94							
Our rewards or returns are comparable to others' gain in the same industry.				0.91	0.93	0.93							
Our rewards or returns reflect well the amount of effort we put in the relationship				0.93	0.90	0.92							
Private Control (adapted from Luo, 2007)	0.95	0.93	0.96				0.97	0.95	0.97	0.88	0.83	0.89	0.891
The supplier (buyer) sometimes creates accidents for its own interest through some very private ways and/or personal interlocks.				0.94	0.90	0.94							
When a conflict or difficulty arises, the supplier (buyer) usually seeks a solution in the best interest of its own (e.g. by exaggerating its losses), not considering us at all.				0.94	0.95	0.95							
The supplier (buyer) chooses not to share information with us on demands, market, competitors, and competitive advantages.				0.95	0.93	0.95							
The supplier (buyer) controls the process in marketing, pricing, operations, and decision making and does not take any suggestion or insight from us.				0.92	0.87	0.93							
Collective Control (adapted from Luo, 2007)	0.91	0.93	0.94				0.94	0.95	0.96	0.79	0.83	0.84	0.934
Both sides work together on establishing new policies and procedures in the face of adversity/challenge.				0.92	0.92	0.94							
Both sides dedicate to establishing new collaboration mechanisms that are suitable for both parties if special problems/needs arise.				0.90	0.90	0.92							
When disagreements arise, both sides work together and try to reach a mutually satisfactory solution.				0.86	0.93	0.91							
Both sides work together on solving problems caused by either side.				0.87	0.90	0.90							
Conflict (adapted from Jap & Ganesan, 2000)	0.97	0.97	0.98				0.98	0.98	0.98	0.93	0.93	0.93	0.876
It is difficult for us to do business with the supplier (buyer).				0.96	0.97	0.96							
The supplier (buyer) makes it hard for us to reach our goals.				0.96	0.97	0.97							
Conflicts of individual personality exist between us and the supplier's (buyer's) representatives.				0.97	0.97	0.97							
We can hardly share our business policy with the supplier (buyer).				0.96	0.95	0.96							
Transaction cost (adapted from Dyer, 1997)	0.93	0.93	0.94				0.95	0.95	0.96	0.82	0.84	0.85	0.891
Our firm uses too many resources and too much time to control the products and production processes of the supplier (buyer).				0.84	0.81	0.84							
It is very difficult to get necessary verification of production performance and production costs data from the supplier (buyer).				0.94	0.95	0.95							
It is very difficult to settle agreements with the supplier (buyer) about specifications of products and services delivered to us.				0.93	0.96	0.96							
It is very time-consuming and difficult to accomplish negotiations between us and the supplier (buyer) about price and payment terms.				0.91	0.93	0.93							
Solidarity (adapted from Kim, 2000)	0.91	0.92	0.93				0.94	0.95	0.95	0.85	0.86	0.87	0.910
A high sense of unity exists between this supplier (buyer) and us.				0.92	0.92	0.92							
We have developed good personal as well as business relationships with this supplier.				0.94	0.94	0.95							
This supplier (buyer) is a very important ally of ours.				0.90	0.93	0.93							
Reciprocity (adapted from Johnson & Sohi, 2001)	0.92	0.92	0.94				0.95	0.94	0.96	0.81	0.81	0.85	0.941
The supplier (buyer) always helps and supports us, and we do likewise.				0.91	0.88	0.92							
In this relationship, both sides feel that one good turn deserves another.				0.91	0.93	0.93							
The supplier (buyer) makes sure that they do their part because they realize we will do ours.				0.90	0.92	0.92							
We feel obliged to do our part extremely well in this relationship because the supplier (buyer) has done their part so well.				0.89	0.87	0.91							
Relationship performance (adapted from Geyskens & Steenkamp, 2000)	0.92	0.89	0.91				0.94	0.92	0.94	0.81	0.75	0.80	0.937
The relationship with this supplier (buyer) has provided us a dominant market position.				0.89	0.81	0.87							
The relationship with this supplier (buyer) has brought us attractive financial gains.				0.90	0.86	0.89							
The relationship with this supplier (buyer) has helped us improve process efficiency.				0.94	0.91	0.93							
This supplier (buyer) provides us with marketing and selling support of high quality.				0.86	0.87	0.87							
Contract completeness (adapted from Luo, 2002)	0.84	0.78	0.84				0.90	0.87	0.90	0.75	0.70	0.76	0.899
Our relationship with this supplier (buyer) is governed by explicitly described and clearly written contract terms				0.84	0.77	0.82							
We included every detail in the contract				0.91	0.91	0.93							
We sign formal contracts to specify each other's obligations.				0.85	0.82	0.86							
Firm type													
1) Wholly State-Owned; 2) Sino-Foreign Joint Venture; 3) Wholly Foreign-Owned; 4) Others_________													
Relationship duration:													
The time span that the relationship between the exchange parties has existed.													

Notes: B represents the buyer data; S represents the supplier data, BS represents the dyadic data; CR = composite reliability; AVE = average variance extracted.

Table 2
Descriptive statistics (N = 216).

Constructs	M	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13
1.Goal difference	4.07	1.20	1.00												
2.Power asymmetry	1.54	1.04	0.04	1.00											
3. Distributive justice	5.51	0.78	−0.03	−0.10	1.00										
4. Private control	3.40	1.24	0.59[a]	0.11[b]	−0.34[a]	1.00									
5. Collective control	5.60	0.80	−0.07	−0.09	0.60[a]	−0.34[a]	1.00								
6. Conflict	2.83	1.40	0.56[a]	0.03	−0.28[a]	0.69[a]	−0.26[a]	1.00							
7. Transaction cost	3.77	1.21	0.62[a]	0.01	−0.15[b]	0.58[a]	−0.12[b]	0.64[a]	1.00						
8. Solidarity	5.73	0.85	−0.00	−0.08	0.62[a]	−0.34[a]	0.70[a]	−0.32[a]	−0.14[b]	1.00					
9. Reciprocity	5.58	0.73	−0.05	−0.08	0.42[a]	−0.26[a]	0.59[a]	−0.18[a]	−0.05	0.50[a]	1.00				
10. Performance	5.55	0.75	−0.10	−0.06	0.53[a]	−0.26[a]	0.67[a]	−0.25[a]	−0.12[b]	0.67[a]	0.57[a]	1.00			
11. Contract completeness	5.33	0.87	−0.10	−0.13[b]	0.21[a]	−0.01	0.28[a]	0.03	0.16[a]	0.25[a]	0.19[a]	0.24[a]	1.00		
12. Firm type	3.33	0.93	0.08	−0.02	0.10	−0.02	0.10	0.05	0.05	0.07	0.04	0.08	−0.00	1.00	
13. Relationship duration	5.44	2.25	−0.00	0.01	−0.06	0.01	−0.01	0.01	0.04	−0.05	−0.05	0.04	0.04	0.01	1.00

Notes: M stands for mean of each variable in the dyadic data; S.D. stands for standard deviation.
[a] Correlation is significant at the 0.01 level (one-tailed).
[b] Correlation is significant at the 0.05 level (one-tailed).

point Likert scales (1 = strongly disagree; 7 = strongly agree), and they (except power asymmetry[6]) were operationalized by taking the means of both sides in the dyadic relationship. For example, relationship performance was measured by averaging the buyer's and the supplier's evaluations of their performance as resulted from the BSR. Following recommendations from Hult et al. (2008) on using multiple types of performance measures to gain a more complete view of performance, four items covering market position, financial gains, process efficiency, and marketing/selling support are included in this study. To check for the mean value approach, we calculated the inter-rater agreement index (Rwg) following James, Demaree, and Wolf (1984). The high Rwg scores for all major variables in our sample (ranging 0.85 to 0.94 in Table 1) indicate a satisfactory agreement or symmetry between two parties (see Kim, 2000), which justifies the mean value approach of operationalizing the constructs in our sample.[7]

In addition to the key constructs, contract completeness, firm type, and relationship duration were chosen as control variables because these two variables affect how Chinese firms view and choose control mechanisms. Contract completeness was measured by three items adapted from Luo (2002), and firm type was a dummy variable. Relationship duration was also included as a control variable due to its influence on relationship quality and performance (Kumar et al., 1995a, 1995b; Palmatier, Dant, Grewal, & Evans, 2006). Relationship duration was measured by the time span of the relationship history between two parties (Palmatier et al., 2006).

The following procedures were conducted to check common method bias and measurement quality. First, we conducted the Harman's one-factor test. An un-rotated factor analysis revealed ten different factors that together accounted for 81% of the variance, and the first factor captured only 13% of the variance. No single factor accounted for most of the variance. Second, we added a general method factor following Podsakoff et al. (2003). The results of the measurement model ($\chi2$ = 6756.35, df = 629, $p < 0.001$, CFI = 0.333, NFI = 0.314, RMSEA = 0.209) indicate that the model fit deteriorates, suggesting common method variance bias may be unlikely. Third, we compared the fit index of two measurement models. One model included traits only, and the other included both traits and a method factor (Huo, Zhao, & Zhou, 2014). The fit of the method factor model was only marginally better than the fit of the traits model (GFI by 0.037; CFI by 0.020, NFI by 0.023; RMSEA by 0.008). These results indicated that the common method factor accounted for only a very small variance. Therefore common method variance bias may not be a big concern in this study.

Next, we assessed reliability using three indices: (1) Cronbach's alphas (between 0.92 and 0.98), indicating good internal consistency; (2) item-to-total correlations (>0.4), implying good external consistency; and (3) composite reliabilities (>0.78) showing high construct reliability. Third, we conducted confirmatory factor analysis (CFA) to assess unidimensionality and convergent validity. The results revealed good model fit for both the supplier data ($\chi^2 = 682$, df = 635, $p < 0.1$, GFI = 0.878, CFI = 0.995, NFI = 0.928, IFI = 0.995, RMSEA = 0.018) and the buyer data ($\chi^2 = 731$, df = 621, $p < 0.001$, GFI = 0.871, CFI = 0.988, NFI = 0.928, IFI = 0.988, RMSEA = 0.028). Average variance extracted (AVE) for all constructs exceeds 0.5, illustrating good convergent validity. Fourth, discriminant validity was evaluated using two-factor CFA models that involved all possible pair of constructs. The chi-square values of all unconstrained models were significantly less than that of the constrained models, providing evidence for discriminant validity. Overall, the constructs in this study have demonstrated high measurement quality.

4. Analysis and results

We examined the levels of private control and collective control in our data. Table 2 shows a moderate level of private control (mean = 3.40) and a relatively higher level of collective control (mean = 5.60). Private control and collective control are negatively correlated at a moderate level ($r = -0.34$). These results suggest the salience of both private control and collective control, as well as the coexistence of both types of control in BSRs. Further, the magnitude (small to medium) and the negative sign of the correlation coefficient suggest that private control and collective control are two distinct control practices and that pure private or collective control is almost nonexistent even if one type of control may be dominant in a certain BSR.

4.1. Results from structural equation modeling

We used structural equation modeling (SEM) to test the proposed model. As shown in Table 3, goal incongruence is positively related to private control ($\beta = 0.651$, $p < 0.01$), while negatively associated with collective control ($\beta = -0.785$, $p < 0.05$). Likewise, power asymmetry is positively related to private control ($\beta = 0.136$, $p < 0.05$) and negatively associated with collective control ($\beta = -0.831$, $p < 0.1$). Distributive justice, however, is negatively related to private control ($\beta = -0.379$, $p < 0.01$) and positively associated with collective control ($\beta = 0.559$, $p < 0.01$). These findings fully support H1, H2, and H3 respectively.

[6] Power asymmetry was not measured directly; instead, power was measured, and power asymmetry was calculated as the absolute difference of power between two parties.
[7] We also tested proposed hypotheses by following Luo (2007) and using the "weighted-mean" approach to operationalize the constructs, and neither the directions nor the significance levels of the hypotheses changed.

Table 3
Hypotheses testing: structural equation model.

Structural antecedent ➔ Action in boundary spanning conduct	β	t-Value	Action in boundary spanning conduct ➔ Governance consequence	β	t-Value
Control variables			Private control		
Contract completeness ➔ Private control	0.10	2.15**	Private control ➔ Conflict	0.66	11.37***
Contract completeness ➔ Collective control	−0.12	−0.97	Private control ➔ Transaction cost	0.42	5.20***
Firm type ➔ Private control	−0.04	−1.02	Private control ➔ Solidarity	−0.11	−2.13**
Firm type ➔ Collective control	0.13	2.68***	Private control ➔ Reciprocity	−0.04	-0.70
Relationship duration ➔ Private control	0.02	0.50			
Relationship duration ➔ Collective control	−0.02	-0.21			
Boundary spanning perspective			Collective control		
Goal difference ➔ Private control	0.65	11.97***	Collective control ➔ Conflict	−0.22	−2.79***
Goal difference ➔ Collective control	−0.80	−2.65**	Collective control ➔ Transaction cost	0.11	1.39
Power asymmetry ➔ Private control	0.14	2.19**	Collective control ➔ Solidarity	0.58	8.72***
Power asymmetry ➔ Collective control	−0.82	−1.74*	Collective control ➔ Reciprocity	0.60	9.10***
Distributive justice ➔ Private control	−0.37	−7.13***			
Distributive justice ➔ Collective control	0.55	7.68***			
R^2: Private control = 0.55, Collective control = −0.56, Conflict = 0.48, Transaction cost = 0.31, Solidarity = 0.55, Reciprocity = 0.38.					

Governance consequence ➔ Performance consequence	β	t-Value
Cost perspective		
Conflict ➔ Relationship performance	0.02	0.42
Transaction cost ➔ Relationship performance	−0.18	−2.76***
Return perspective		
Solidarity ➔ Relationship performance	0.52	8.03***
Reciprocity ➔ Relationship performance	0.33	5.73***
R^2: Relationship performance = 0.58.		
Model Fit: χ^2 = 703, df = 604, p < 0.01, GFI = 0.88, CFI = 0.99, NFI = 0.93, IFI = 0.99, RMSEA = 0.02		

* p < 0.1.
** p < 0.05.
*** p < 0.01.

As expected, for private control, significant and positive coefficients are obtained between private control and conflict ($\beta = 0.668$, $p < 0.01$), and between private control and transaction cost ($\beta = 0.419$, $p < 0.01$). Similarly, a significant and negative coefficient is achieved between private control and solidarity ($\beta = -0.118$, $p < 0.05$). However, the proposed negative link between private control and reciprocity is non-significant ($\beta = -0.043$, n.s.). For collective control, although a significant negative relationship is found between collective control and conflict ($\beta = -0.220$, $p < 0.01$), the coefficient between collective control and transaction cost is non-significant ($\beta = 0.108$, n.s.). Furthermore, solidarity ($\beta = 0.581$, $p < 0.01$) and reciprocity ($\beta = 0.608$, $p < 0.01$) are both positively related to collective control. As for the relationship between mid-range consequences and ultimate consequences, dyadic relationship performance is negatively associated with transaction cost ($\beta = -0.180$, $p < 0.01$), but not with conflict ($\beta = 0.025$, n.s.), and is positively related to both relationship solidarity ($\beta = 0.521$, $p < 0.01$) and reciprocity ($\beta = 0.335$, $p < 0.01$). Thus, these results lend partial support to H4 and H5.

4.2. Results validation from bootstrap testing

We validated H4 and H5 using bootstrapping, a method recommended by Bollen and Stine (1990) for the analysis of direct and indirect effects in mediation. Regarding the mediating role of conflict, the bias-corrected bootstrap estimates (95% CI) of a1 (0.62, 0.94) and a2 (−0.56, 0.00) exclude zero (see Table 4), indicating that private control is positively while collective control is negatively associated with conflict. The estimate of b1 (−0.10, 0.12) includes zero, which means that conflict is not related to relationship performance. In addition, the estimates of a1 ∗ b1 (−0.08, 0.10) and a2 ∗ b2 (−0.05, 0.02) both include zero, showing that the indirect effects of private control and collective control on relationship performance through conflict

Table 4
Hypothesis testing: bootstrap results (N = 1000).

	95% CI							
	Conflict		Transaction cost		Solidarity		Reciprocity	
Effect	Bootstrap percentile	Bias-corrected bootstrap	Bootstrap Percentile	Bias-corrected bootstrap	Bootstrap percentile	Bias-corrected bootstrap	Bootstrap percentile	Bias-corrected bootstrap
Private control								
a1	(0.61, 0.94)	(0.62, 0.94)	(0.51, 0.79)	(0.51, 0.80)	(−0.16, 0.00)	(−0.16, 0.00)	(−0.11, 0.04)	(−0.11, 0.04)
b1	(−0.10, 0.12)	(−0.10, 0.12)	(−0.15, 0.00)	(−0.15, 0.00)	(0.14, 0.52)	(0.16, 0.54)	(0.08, 0.39)	(0.09, 0.40)
c1′	(−0.13, 0.08)	(−0.12, 0.09)	(−0.03, 0.13)	(−0.03, 0.13)	(−0.04, 0.07)	(−0.04, 0.07)	(−0.05, 0.05)	(−0.05, 0.04)
a1 ∗ b1	(−0.08, 0.10)	(−0.08, 0.10)	(−0.10, 0.00)	(−0.11, 0.00)	(−0.05, 0.00)	(−0.06, 0.00)	(−0.03, 0.01)	(−0.03, 0.00)
Collective control								
a2	(−0.56, 0.00)	(−0.56, 0.00)	(−0.01, 0.37)	(−0.02, 0.35)	(0.57, 0.88)	(0.57, 0.88)	(0.45, 0.75)	(0.45, 0.76)
b2	(−0.10, 0.12)	(−0.10, 0.12)	(−0.15, 0.00)	(−0.15, 0.00)	(0.14, 0.52)	(0.16, 0.54)	(0.08, 0.39)	(0.09, 0.40)
c2′	(0.47, 0.85)	(0.48, 0.85)	(0.52, 0.86)	(0.54, 0.87)	(0.22, 0.63)	(0.21, 0.61)	(0.35, 0.72)	(0.35, 0.73)
a2 ∗ b2	(−0.05, 0.02)	(−0.05, 0.02)	(−0.04, 0.00)	(−0.04, 0.00)	(0.11, 0.38)	(0.13, 0.42)	(0.05, 0.24)	(0.05, 0.25)

Note: Using the private control–conflict–satisfaction path as an illustration: a1 denotes the effect of private control on conflict; b1 denotes the effect of conflict on relationship performance; c1' denotes the direct effect of private control on relationship performance; a1 ∗ b1 denotes the indirect effect of private control on relationship performance through conflict.

respectively are not significantly different from zero. Therefore, conflict is not a mediator between both types of controls and relationship performance. For transaction cost, the bias-corrected bootstrap estimate (95% CI) of a1 (0.51, 0.80) excludes zero while a2 (−0.02, 0.35) includes zero, indicating that private control is positively associated with transaction cost while collective control is not. The estimate of b1 (−0.15, 0.00) also excludes zero, which means that transaction cost is related to relationship performance. In addition, the estimate of a1 ∗ b1 (−0.11, 0.00) excludes zero, showing that the indirect effect of private control on relationship performance is significantly different from zero. Therefore, transaction cost is a mediator between private control and relationship performance, but not a mediator between collective control and relationship performance. Thus, H4 is partially confirmed.

As to solidarity, the bias-corrected bootstrap estimates (95% CI) of a1 (−0.16, 0.00) and a2 (0.57, 0.88) exclude zero, indicating that both private control and collective control are significantly associated with solidarity. The estimate of b1 (0.16, 0.54) also excludes zero, which means that solidarity is related to relationship performance. In addition, the estimates of a1 ∗ b1 (−0.06, 0.00) and a2 ∗ b2 (0.13, 0.42) both exclude zero, showing that the indirect effects of private control and collective control on relationship performance through solidarity respectively are significantly different from zero. Thus, solidarity is a mediator between both types of controls and relationship performance. Lastly, the bias-corrected bootstrap estimate (95% CI) of a1 (−0.11, 0.04) includes while a2 (0.45, 0.76) for reciprocity excludes zero, indicating that collective control is significantly related to reciprocity while private control is not. The estimate of b2 (0.09, 0.40) also exclude zero, which means that reciprocity is related to relationship performance. In addition, the estimate of a2 ∗ b2 (0.05, 0.25) excludes zero, indicating that the effect of collective control on relationship performance through reciprocity is different from zero, and supporting that reciprocity is a mediator between collective control and relationship performance. Therefore, H5 is also partially confirmed.

5. Discussion and conclusion

Extending the private/collective control scheme to BSRs, this study examines the boundary-spanning interfaces in BSRs that may trigger exchange parties' private and collective controls as well as the various governance and performance outcomes of private and collective controls. The results support our theorized view of (1) the coexistence of private and collective controls in a BSR; (2) controls in a BSR as both endogenous and associated with relationship structure; and (3) controls exerting multifaceted consequences in a BSR. We believe that an examination of private and collective controls in BSRs helps deepen our understanding of interorganizational governance and add explanatory power to the efficacy of controls in boundary spanning conduct in BSRs in supply chain/channel management.

5.1. Theoretical implications

This study provides several implications for interorganizational governance research in general and BSRs research in particular. First, by applying an alternative control scheme to BSRs, this study contributes to our understanding of the control/governance issues in BSRs research by revealing that exchange parties may engage in private and collective control practices because the buyer and the supplier are both dependent and interdependent in a BSR. In examining parties' boundary spanning conduct in BSRs, we differentiate between private control and collective control, showing that private control involves unilateral actions for the purpose of self-gain for the individual party, and that collective control entails bilateral actions for the sake of mutual gain for both parties. The results suggest that private control and collective control are two distinct types of behavioral control that prevail in and are important for BSRs, which echoes Kumar's (2005) and Das and Teng's (1998) views that partnerships have both cooperative (positive-sum) and competitive (zero-sum) dimensions. In other words, certain degrees of exchange parties' unilateral and bilateral actions are simultaneously present; and it is rare to have pure private or collective control in a BSR.

Second, based on boundary spanning theory, we theorize that control practices in a BSR are triggered by boundary spanning interfaces. By demonstrating that private and collective controls are influenced by three structural interfaces, this study presents that private and collective controls are not exogenous, but rather determined by the characteristics of the BSR. Drawing from the BSRs literature, this study identified three interface conditions. The first one, goal incongruence, is based on value position in a BSR. Goal incongruence constrains collective control and stimulates private control because misfit configurations in organizational goals can cause interest diversity and identity separateness, encouraging parties' desire for self-interest and curbing their enthusiasm for joint actions. This finding is consistent with prior studies submitting that goal incongruence stimulates unilateral opportunistic behavior whereas goal congruence promotes cooperation in BSRs (Jap, 2001; Rossetti & Choi, 2008). As well, this study extends Luo's (2007) finding of the importance of goal congruence (or the consequence of goal incongruence) in horizontal interorganizational partnerships to include vertical BSRs in supply chains and distribution channels.

The second interface condition, power asymmetry, works similarly to goal incongruence in that difference between exchange parties in bargaining power is negatively related to collective control and positively associated with private control. Previous studies link power asymmetry with conflict in a BSR because the more powerful party has less motivation to avoid conflict since it perceives retaliation from the weaker party to be less likely and less damaging (Kumar et al., 1995a, 1995b) and because the less powerful party is more likely to engage in a preemptive strike or rebellion against the more powerful party's domination since it perceives exploitation as inevitable (Lawler, Ford, & Blegen, 1988). By introducing private control and collective control in the BSR context, our study further illuminates the underlying mechanisms of private control and collective control that link power asymmetry and conflict.

In contrast, the third interface condition, distributive justice, is found to have a positive impact on collective control and a negative effect on private control. Existing studies on boundary spanning typically focus on interactions on resources, goals, power and culture between firms (Bradach & Eccles, 1989). We believe distributive justice as an important boundary spanning link that influences the extent of both collective and private controls. Equity theory treats justice (or fairness) as a fundamental foundation for all types of social and economic relationships, especially those long-term relationships (Greenberg, 1993). As the first and most important dimension of justice, distributive justice has received extensive attention. The underlying logic behind the importance of distributive justice is that compared with the exact amount of outcomes, people care more about whether the outcomes are fairly distributed (Homans, 1961). When the level of distributive justice is high, both parties perceive that their investments are adequately recognized and will be rewarded proportionally; thus, they worry less about the other party's exploitation and opportunism in the relationship (Griffith et al., 2006). Accordingly, they are more willing to work together, share knowledge, and make mutual investment to maximize their joint gains while at the same time refrain from engaging in unilateral self-interest-seeking behaviors in boundary spanning activities that would harm the relationship (Liu et al., 2012).

Third, regarding the consequences of private and collective controls, existing studies mainly address the direct consequences of private and collective control (Luo, 2007; Luo et al., 2008). Our study extends such research to further examine the impacts of private and collective controls on the ultimate relationship performance, knowing that the performance implications of private and collective controls can provide firms with direct and clear guidelines in control choice and resource allocation. In addition, distinguishing the mid-range governance consequences from the ultimate performance adds insight to interorganizational relationship management research because a study of multi-layer consequences helps identify the specific paths through which private and collective controls exert their effects on the overall relationship performance. Our results indicate that, in a BSR, it is through reducing governance cost (i.e., conflict, transaction cost) and increasing governance return (i.e., solidarity, reciprocity) that private and collective controls affect the overall relationship performance. Specifically, private control gradually destructs relationship performance by introducing a perspective of isolation that centers on a plague of irreconcilable discord and continuous suspicion, while collective control helps improve relationship performance through fostering a sense of unity that rests on the entangling of mutual interests, inner cohesion, and reciprocal reinforcement.

Fourth, departing from many previous studies that primarily take on one-side perspective in BSRs (e.g., Griffith et al., 2006; Heide et al., 2007; Jaworski, 1988), this study approaches control-based governance from a dyadic perspective, probing control mechanisms from both the buyer and the supplier perspective. Traditionally, the single firm has been the unit of analysis in management research (Wright & Lockett, 2003), but in our view, adopting the dyad—rather than the single firm—as the unit of analysis and examining the relationship-specific structural conditions and performance consequences of control, seem more appropriate in interorganizational collaborations because the ultimate goal of an cooperative relationship is that both parties should benefit from and thrive in the relationship (Dyer, 1997).

5.2. Managerial implications

Practitioners need to be aware that some degrees of private control and collective control may be simultaneously present because the buyer and the supplier are both dependent and interdependent, and thus partners may simultaneously engage in both private and collective controls of boundary spanning behaviors in a BSR. Further, because private control hinders whereas collective control helps achieve favorable performance from both parties (i.e., overall relationship performance), managers need to recognize what factors activate collective control and private control of boundary spanning behaviors and what impacts these control practices bring to the BSR. Our results provide some initial guidance for managers in this regard.

Among the three interparty structural conditions we examined, goal incongruence and power asymmetry between two parties lead to private control. Therefore, for managers who seek to restrain partners' private control during boundary spanning interactions in supply chains or distribution channels, an advice is to choose a supply chain/channel partner with similar organizational goals and equal bargaining power. This is because partnering with these like organizations can reduce structural looseness, therefore inducing more collective control and less private control in boundary spanning conduct. Our results also suggest that established distributive justice exerts a positive effect on collective control and a negative effect on private control. This implies that the more balanced the outcome distribution is, the more likely organizations will choose collective control in interparty boundary spanning conduct. Thus, a suggestion for parties in BSRs would be to focus on achieving perceived distributive justice by ensuring that both parties share the perception that outcomes are distributed fairly, and that each party receives fair returns and profits. For instance, rather than squeezing suppliers for continuous price concessions, the buyer (such as Walmart at present, and Toyota in the 1970s and 1980s, as reported in Day, 1990 and Langfield-Smith & Greenwood, 1998) may determine purchasing prices together with the supplier at levels that are perceived as fair by both parties, and that ensure adequate profits for the supplier.

Regarding the various effects of private and collective controls, our results show that private control induces conflict and transaction cost whereas collective control provokes solidarity and reciprocity which in turn affect relationship performance. Solidarity and reciprocity in a BSR improve while transaction cost subdues relationship performance. Recall that relationship performance concerns both parties' performance resulted from the relationship. Only when both parties simultaneously exhibit more collective control and less private control can relationship performance be improved. Thus, for partners who are concerned about both parties performing well in interorganizational collaboration, engaging in collective control as well as curbing private collect is the key to effective relationship management.

5.3. Limitations and future research

This study has several limitations that merit further investigation. First, the results suggest that both private and collective controls present in any BSRs; however, this study did not reveal the patterns or the dynamics of the two types of controls. As mentioned, private control and collective control are negatively correlated at a moderate level. Future research may probe into the detail of the control patterns in different types of BSRs under different types of governance. Further, BSRs are not static; control practices may change and evolve over the different relationship stages (Claycomb & Frankwick, 2010). It is possible that partners may execute different control configurations during different stages in the relationship lifecycle (Jap & Ganesan, 2000). Future studies may delve into case studies to provide insight into the configurations and the dynamics of private and collective controls under different governance types and in different relationship stages (see Langfield-Smith & Greenwood, 1998; Majchrzak et al., 2015; Varoutsa & Scapens, 2015).

Second, we identified three structural conditions (goal incongruence, power asymmetry, and distributive justice) in a BSR based on the boundary spanning structural interface. By no means has this represented an exhausted list of antecedents. Following Brown et al. (1995), the power construct in our study mainly captures the measure of coercive power because, as Brown et al. (1995) indicates, coercive power is more effective in guiding behavior than non-coercive power. Another reason for focusing on coercive power is that the research setting of this study—the Chinese household appliances industry— is not a knowledge intensive industry. The reinforcements of non-coercive power, such as knowledge or expert, are not dominant BSRs in this industry. However, the influential effect of knowledge or expert power on exchange parties' control practices may be salient in high-technology industries. Future studies may include non-coercive power in the model and examine the effect of the asymmetry in non-coercive power such as expert power and reward power on private and collective controls[8] in BSRs, using data from other industries. In addition to the structural interfaces in a BSR, other forces internal and external to BSRs may also prompt private and collective controls. For example, interparty relationship specific investment derived from transaction cost economics (Williamson, 1979) may trigger private control and collective control respectively; mutual trust and commitment based on trust-commitment theory (Morgan & Hunt, 1994) may promote collective control; and competitive intensity in the external environment may encourage private control. Future studies may further look at other antecedents (e.g., organizational cultural, Das & Teng, 1998) to form a more complete picture of private and collective controls.

Third, this study collected data from the home appliance industry in China, an emerging economy, to test the proposed model. Such a one-industry research setting, while excluding the industry-level variance, may limit the generalization of results to other industries (particularly with the aforementioned exercising of coercive and non-coercive power) or countries. Follow-up studies attempting to verify our model and results would be a valuable contribution, using data from other industries and countries. Conducting comparative research across emerging economies and developed countries (the same industry) could also offer significant insights (Bensaou & Venkatraman, 1995). Further, in an international buyer-supplier setting, an international BSR may be

[8] We thank one anonymous reviewer for this suggestion.

subject to culture differences and external institutional forces when the buyer and the supplier are from different countries (Wright et al., 2005). Future studies in the global context may consider incorporating into the model such external boundary conditions as cultural distance and other institutional-level differences, either as antecedents to controls or as moderators interacting with relationship structure variables such as power asymmetry.

Fourth, this study focuses on manufacturer-distributor dyads. Distribution channels and supply chains nowadays, however, are constructed in the form of 'netchains' (e.g., Toyota's network of suppliers and Pepsi's network of distributors) consisting of layers of horizontally-linked suppliers (or distributors), which are also associated with the same buyer (or manufacturer) (Larson, 1992). Some interesting questions worth investigating would be how private and collective controls in vertical links affect horizontal links, and how horizontal ties affect the control mechanisms in vertical relationships.

Finally, although our proposed model implies causal relationships, the data collected is cross-sectional. Future studies need to collect longitudinal data to verify causality implied in the model. A longitudinal perspective will also help reveal the dynamics of control practices in the abovementioned different relationship stages, providing a more holistic picture of the co-existence and co-evolvement of private and collective controls in exchange relationships.

Acknowledgment

The authors are grateful to the editor and anonymous reviewers for their constructive suggestions. We acknowledge the financial support of the National Natural Science Foundation of China (71572109; 71502066) and Program for Changjiang Scholars and Innovative Research Team in University (IRT13030).

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